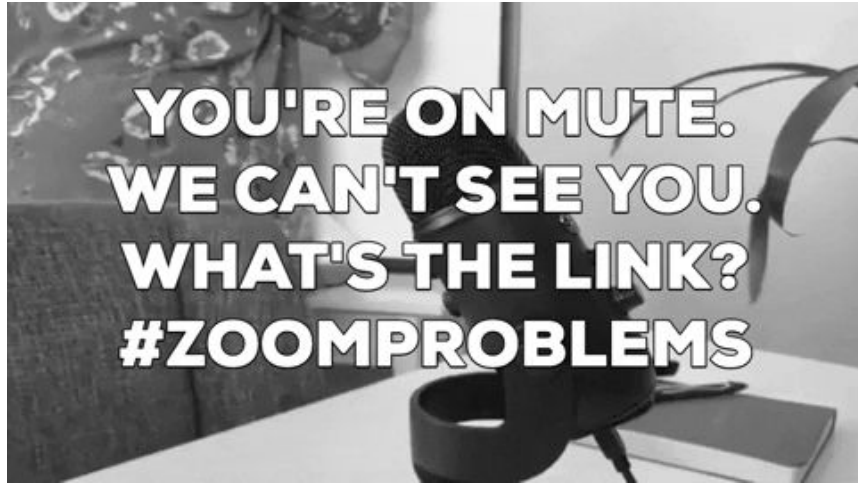


COMS W4156 Advanced Software Engineering (ASE)

September 7, 2023



Repeated from Tuesday: Important Notice



All of my classes are held on Zoom as part of my ADA accommodations (the course counts as "in person" for ISSO purposes)

If you do not want to take this course on Zoom, then drop the course

Prof. Yang's [W4152 Engineering Software as a Service](#) is an acceptable substitute for most students who are required to take this course

Lecture Notes linked on [Courseworks Calendar](#)

'Live' links into google docs, so possibly changing up to and even during class

Pdfs will be posted on [Courseworks->Files->lecture notes](#) later

Questions and Comments during Class

[Zoom](#), either voice (raise hand) or chat (type whenever you like)

Questions and Comments outside of Class

[Ed Discussion](#) (supports anonymous and private posts)



Are you on the Waitlist?

Student Services Online

► **Wait List**

If you are a Columbia or Barnard undergraduate, I am waiting for your response to the request I've sent several times asking for when you took COMS W3157 Advanced Programming and what was your grade. I will not approve you to enroll in the course until I hear from you (send email to kaiser+4156@cs.columbia.edu).

If you are a CS or CE MS student, you have already been approved unless you joined the waitlist very recently. I'll check the waitlist again later today.

If you are on the waitlist but neither of the above: I'll start admitting non-majors from the waitlist after today's class to fill the space still available.

Still Repeating from Tuesday: Team Project

Ideally 4 students per team, 5 students allowed (no teams of 1, 2, 3, 6, 7, ...)

Students arrange their own teams, use [Ed Discussion](#) (and/or anything else you want) to find team members

Read [Coping with Hitchhikers and Couch Potatoes on Teams](#) *before* forming team

Every team will have a Mentor who is either an IA or a Leader for this course

Team projects will develop and test a project of their own choice, within constraints



Team Project Constraints

Both the service and the client app must be developed in Java, C++ or Javascript (plus XML/HTML/CSS)

Python is not allowed

The client app must target an acceptable user group that is not university students, staff or faculty (except for individuals who happen to also be members of an acceptable user group)

It should not be possible for someone reading only the service's code (including comments) to guess what the client app does or who is its target user group

The client depends on the service but the service must not depend on the client

Dependencies on third-party software artifacts such as other services, libraries, frameworks, and AI-generated code is ok but your use must be clearly documented



CONSTRAINTS
FORCE
creativity

Acceptable Target Users

- Elderly (defined for this purpose as age 75 or over)
- Eldercare workers (caregivers for elderly, often non-English speakers)
- Disabled or impaired (e.g., vision, hearing, speech, motor, cognitive)
- Healthcare workers (doctors, medical technicians, nurses, aides, etc.)
- PreK-2 (early childhood) school teachers
- First responders (police, fire, EMT)
- Maybe others with "societal benefits" - ask me *in advance*

Remember: Your client needs to know about your target user group but your service does not, your service should be useful for a variety of client apps targeting different user groups



Team Project Assignments

TEAM FORMATION

PRELIMINARY PROJECT
PROPOSAL

(Team meets with Mentor after
preliminary proposal to receive
feedback towards revised proposal)

REVISED PROJECT PROPOSAL



FIRST ITERATION

FIRST ITERATION DEMO
(for Mentor)

SECOND ITERATION

SECOND ITERATION DEMO
(for Mentor)

DEMO DAY

(for me and any students in class who
want to watch other teams' demos)

“Differential Grading”



Default is for all members of the same team to get the same grade for the project

But every year there are some students who do not do their fair share

Such students will receive a lower grade than their teammates, possibly a much lower grade, based on input from their other team members and their team mentor

Differential grading is applied overall at end of semester, not per assignment

Read [Coping with Hitchhikers and Couch Potatoes on Teams](#) *before* forming team!

Catching Up



That's all I'm going to repeat from Tuesday's class

If you missed Tuesday's class, watch the recording in [courseworks video library](#)

But do not rely too much on recordings

After the program change period ends, we will start tracking attendance

I will also call on random students to respond to material during class sessions

So please pay attention, do not just start up zoom and walk away

And make sure your zoom name matches your coursework's name to get credit

Agenda

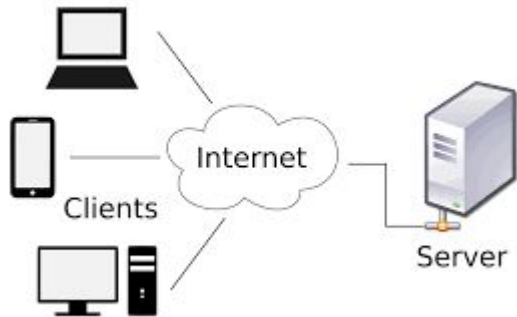
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5. What is the Difference between a Service and a Library?
6. What is the Difference between a Service and an App Backend?
7. What is the Difference between a Service and a Microservice?



What is the Difference between a Service and a Server?

Answer 1. Nothing - these terms are sometimes used interchangeably

Answer 2. Software vs. Hardware - sometimes the term 'server' refers to a physical machine, or to a [datacenter](#) (large collection of physical machines), rather than to software. Distinction usually clear from context



What is the Difference between a Service and a Server? 2

Answer 3. Scale - Servers are 'big' services or collections of services

Answer 4. Storage - Some services are stateless, or store transient state in memory or on disk only while corresponding client sessions are in progress, whereas others store persistent state (files and databases) between sessions. Servers almost always store persistent state and serve something from disk, such as web pages



What is the Difference between a Service and a Web Service?

Not much

Once upon a time, "web service" referred to a service whose API was defined in the Web Services Description Language (WSDL) notation. The [WSDL spec](#) was last updated in 2007 and is still used in the same sense that COBOL is still used

But today, "web service" can mean any service accessed over the internet



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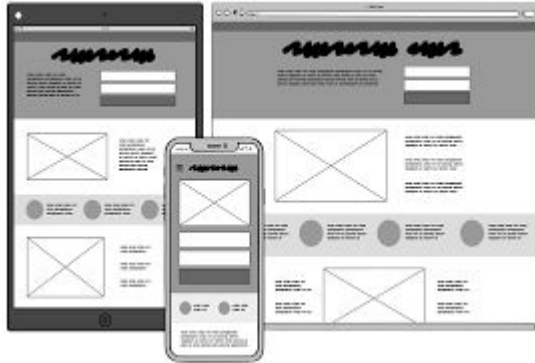


What is an App?

Software that does something for users

The developers of an app design a UI for users, often with a diagramming tool like draw.io

The internal design of the app code defines a set of interacting software units that together implement the UI



What is a Service?

Software that does something for other software

The developers of a service design an API for clients, often with a specification tool like [OpenAPI](https://openapi.org/)

The internal design of the service code defines a set of interacting software units that together implement the API



What is the Difference between a Service and an App?

If you ask google search, you will find a variety of contradictory answers



For the purposes of this course, an app (or application) is used by human users via a user interface (UI) whereas a service is used by code via an application programming interface (API)

Please do not say something like “our team’s service is an app that does xxx”, “the user does xxx with our service”, or “the user uses our service to do xxx”

If a human user is using it, then it is an app not a service

Instead say “our team’s service does xxx”, “the client does xxx with our service”, or “the client uses our service to do xxx”

Are there any Exceptions?



When I say human users, I mean human end-users - not human developers or administrators

A service may have a command line interface (CLI) or even a graphical user interface (GUI) for administrators to start/stop the service and perform other administrative actions, but it is not available to end-users

Developers test APIs with tools that mimic client code

- ★ For example, [Postman](#) and [SoapUI](#)

Developers test UIs with tools that mimic human end-users

- ★ For example, [Selenium](#) and [Cypress](#)



What about Software as a Service?

Software as a Service ([SaaS](#)) performs the service of supplying users with apps that are hosted on the cloud rather than being installed on the users' machines

But the human user still interacts with each app via a UI and code (that implements an app) interacts with the service via an API

A human user can use an app but cannot use a service, only clients can use a service



Non-SaaS Application



Application logic runs
on user's computer

SaaS Application



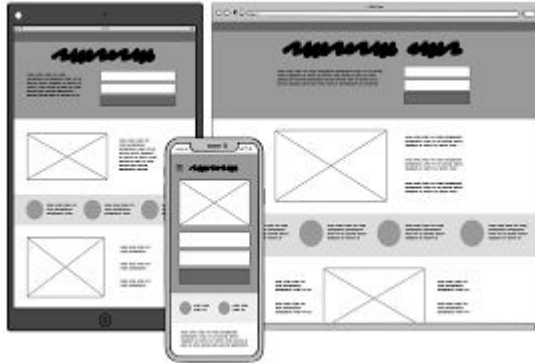
Application logic runs
in the cloud

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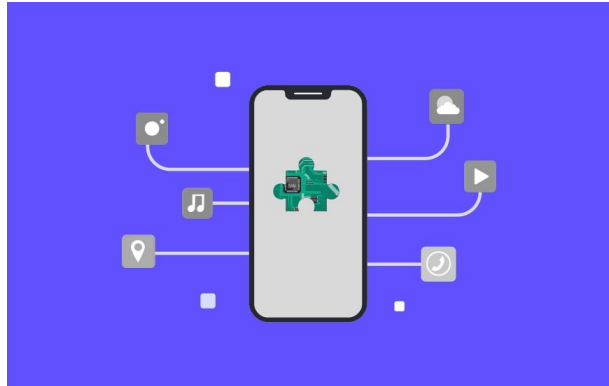
SaaS Application



Application logic runs
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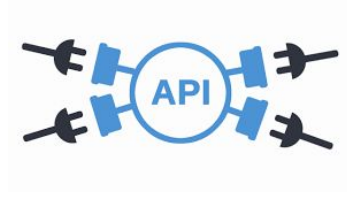


What is an API?

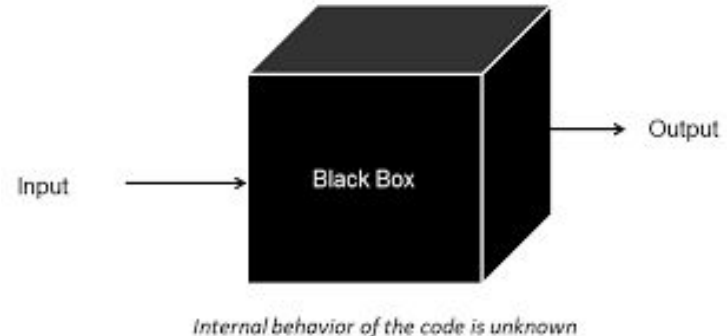
APIs are the middlemen allowing software to communicate with other software

An API consists of one or more exported entry points:

- synchronous method/function calls (from the client to the service)
- [asynchronous messages](#) (from the client to the service)
- [callbacks](#) (from the service to a method/function registered by the client)
- [references to data](#) (from the client to a data service)



Typical APIs are 'blackbox': the developer's code uses the interface provided by the box but the developer should not (or can not) peek inside the box to see how it works



Libraries have APIs

Services are not the only software units with APIs, libraries and [frameworks](#) also provide APIs. If you've written a program that does pretty much anything, you've already used APIs

Most programming languages have “standard libraries” that come with the language or the platform (e.g., JVM, Linux), for I/O, string processing, math, system calls



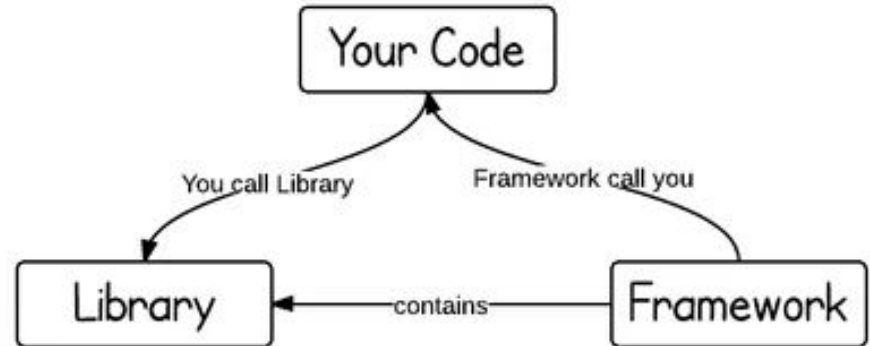
'Third-party' libraries are downloaded manually or via a build tool / package manager ecosystem (e.g., [Maven Central](#), [npm](#))

Who are the three parties? 1. Your team 2. Your customers and users 3. Any other source of software besides your team (or your organization)

What is the Difference between a Service and a Library?

A library is shared code that the developer compiles or bundles into their app to run in the same process

A service is shared code running in another process, often on another machine, that the developer accesses from their app

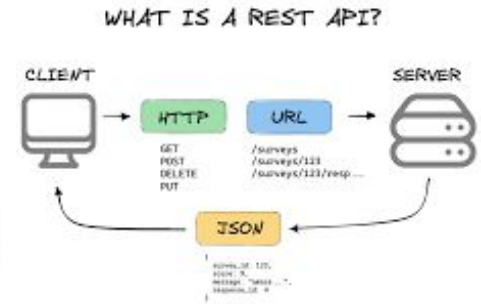
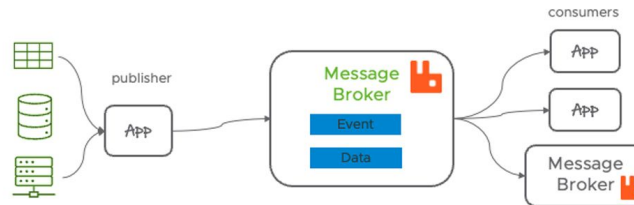
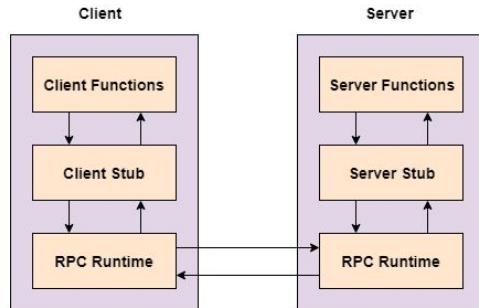


Calls to Library functions vs. Calls to Service functions

The developer's code calls a library function just like any other function

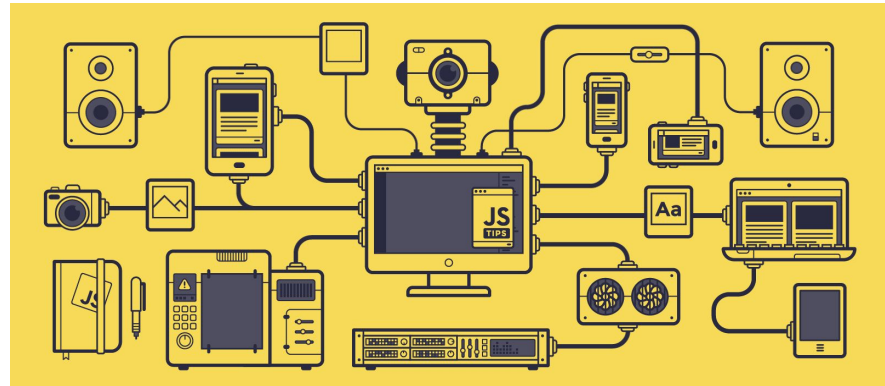
The developer's code requires additional infrastructure to call a service function

- Remote procedure call, e.g., [gRPC](#)
- Message broker, e.g., [RabbitMQ](#)
- REpresentational State Transfer ([REST](#)), e.g., [Spring](#)



Agenda

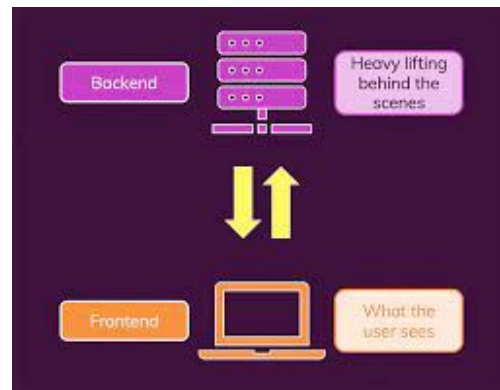
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What is the Difference between a Service and an App Backend?

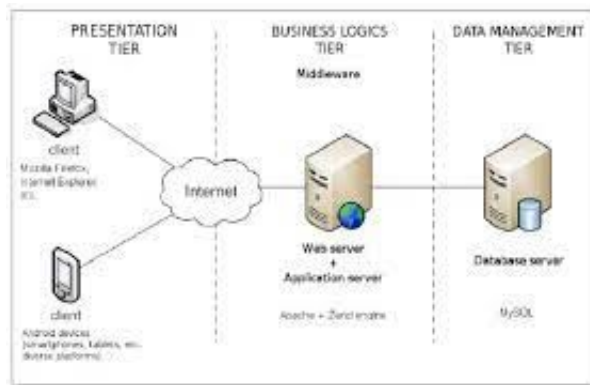
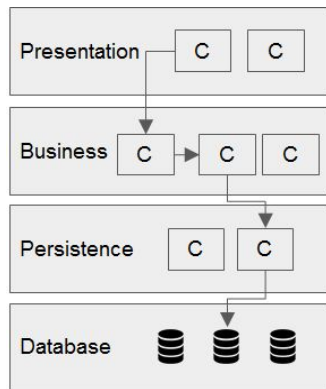
Apps often consist of a front-end running on the user's device and a back-end running in the cloud

This sounds like client/server, but is typically implemented with three or more 'tiers'. The back-end is everything except the presentation layer (or tier)



Is an app's back-end a service? 😞 No 😞

If the back-end is part of this one app, and only this one app, then it's not what I mean by a service. Services should be usable by (and useful to) many different apps

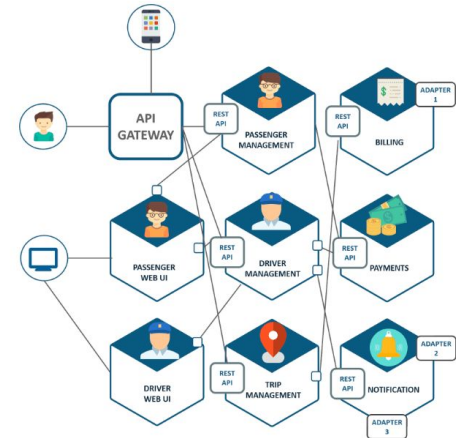
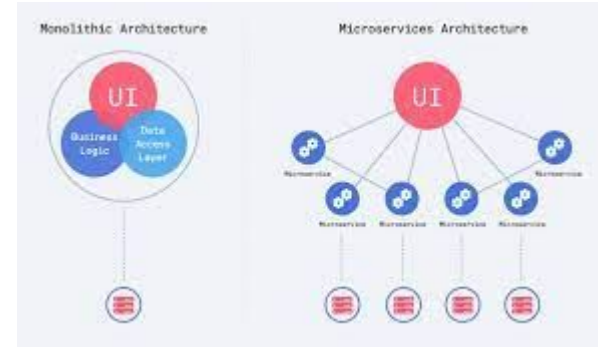


What is the Difference between a Service and a Microservice?

But a single app's back-end can be organized into [microservices](#)

Microservices refers to a software architecture pattern (or [architectural style](#)) where an app is built as collection of small, independent, and loosely coupled service-like pieces that work together to provide the complete functionality

Each microservice can be developed by a separate team using its own choice of programming language and development technology, and deployed independently, allowing for greater flexibility and scalability in large-scale applications



Next Week

Working in Teams

Finding Bugs

There will be (much) more on APIs later on



Are you on the Waitlist?

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Ask Me Anything